CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

November 2012

EC203 Solid State Electronics

Date: 29.11.2012, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a)
1. At absolute temperature, the conduction band of semiconductor is totally ________. **[05]**
2. With the rise in temperature, the electrons of the valance band of a semiconductor are lifted to ___________ band.
3. To obtain p type semiconductor, __________ impurity is added to a pure semiconductor.
4. The addition of pentavalent impurity to a semiconductor creates _________.
5. A semiconductor has _________ temperature coefficient of resistance.

(b) What is the importance of peak inverse voltage? **[02]**

Q - 2 (a) A full wave centre tap rectifier uses two crystal diodes each having a forward resistance of 25 ohm. The r.m.s. value of secondary voltage fed between centre tap to each end of secondary is 48V and the load resistance is 1Kohm.Find i)d.c. output voltage ii) d.c. output power iii)rectification efficiency iv) peak inverse voltage. **[07]**

(b) Write a short note on a maximum voltage rating for transistor and derive the relation between α and β. Also find base current for α=0.99, I_C = 6 mA and I_{CBO} = 10 μA. **[07]**

OR

Q - 2 (a) Derive an expression for the efficiency of a half wave rectifier. **[07]**

(b) Explain the output characteristics of common emitter transistor. Label the three operating regions over it. **[07]**

Q - 3 (a) Derive the general formula for (i) input impedance (ii) current gain (iii) voltage gain in terms of h parameter and the load for a common emitter transistor amplifier. **[07]**

(b) Draw the 'h' parameter equivalent circuit for a typical common emitter amplifier and derive expression for A_i ,R_i ,A_v and $R_{o'}$. **[07]**

OR

Q - 3 (a) Compare Common emitter, common base and common collector transistor configuration. **[07]**

(b) The following readings were obtained experimentally from a JFET: [07]

Determine (i)a.c.drain resistance (ii)transconductance(iii)amplification factor

V_{GS}	0V	0V	-0.2V
V_{DS}	7V	15V	15V
I_D	10mA	10.25mA	9.65mA

SECTION – II

Q - 4 (a) What is the difference between a JFET and a bipolar transistor? Explain the construction and working of a JFET. [07]

Q - 5 (a) Draw the Generalized Hybrid Model for transistor amplifier and derive all the parameters. [07]

(b) Write a note on hybrid π capacitances. [07]

OR

Q - 5 (a) Explain the transistor emitter follower response at high frequency in detail. [07]

(b) Write a note on hybrid π conductance. [07]

Q - 6 (a) Draw the circuit of an R-C (Resistance – Capacitance) coupled Transistor Amplifier. Draw its gain v/s frequency characteristics and indicate cut off frequencies and bandwidth. [07]

(b) Explain high frequency response of multistage Common emitter transistor amplifier. [07]

OR

Q - 6 (a) Explain the following terms of multistage amplifier. [07]

(i)frequency response (ii)Decibel gain (iii)Bandwidth gain

Also sketch its response.

(b) Compare different type of multistage amplifier with its parameters. [07]

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